## Question ID: 85939da5

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Inference from sample statistics and margin of error | Hard |

**Question**

| Texting behavior | Talks on cell phone daily | Does not talk on cell phone daily | Total |
|---|---|---|---|
| **Light** | 110 | 146 | 256 |
| **Medium** | 139 | 164 | 303 |
| **Heavy** | 166 | 74 | 240 |
| **Total** | 415 | 384 | 799 |

In a study of cell phone use, 799 randomly selected US teens were asked how often they talked on a cell phone and about their texting behavior. The data are summarized in the table above. Based on the data from the study, an estimate of the percent of US teens who are heavy texters is 30% and the associated margin of error is 3%. Which of the following is a correct statement based on the given margin of error?

**Answer**

A. Approximately 3% of the teens in the study who are classified as heavy texters are not really heavy texters.

B. It is not possible that the percent of all US teens who are heavy texters is less than 27%.

C. The percent of all US teens who are heavy texters is 33%.

D. It is doubtful that the percent of all US teens who are heavy texters is 35%.

**Correct Answer: D**

**Rationale**

Choice D is correct. The given margin of error of 3% indicates that the actual percent of all US teens who are heavy texters is likely within 3% of the estimate of 30%, or between 27% and 33%. Therefore, it is unlikely, or doubtful, that the percent of all US teens who are heavy texters would be 35%.

Choice A is incorrect. The margin of error doesn't provide any information about the accuracy of reporting in the study. Choice B is incorrect. Based on the estimate and given margin of error, it is unlikely that the percent of all US teens who are heavy texters would be less than 27%, but it is possible. Choice C is incorrect. While the percent of all US teens who are heavy texters is likely between 27% and 33%, any value within this interval is equally likely. We can't be certain that the value is exactly 33%.

## Question ID: 954943a4

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Percentages | Hard |

**Question**

Jennifer bought a box of Crunchy Grain cereal. The nutrition facts on the box state that a serving size of the cereal is $\frac{3}{4}$ cup and provides 210 calories, 50 of which are calories from fat. In addition, each serving of the cereal provides 180 milligrams of potassium, which is 5% of the daily allowance for adults. If p percent of an adult's daily allowance of potassium is provided by x servings of Crunchy Grain cereal per day, which of the following expresses p in terms of x ?

**Answer**

A. $p = 0.5x$

B. $p = 5x$

C. $p = (0.05)^x$

D. $p = (1.05)^x$

**Correct Answer: B**

**Rationale**

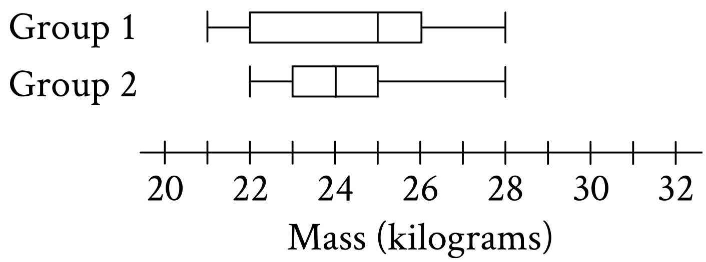
Choice B is correct. It's given that each serving of Crunchy Grain cereal provides 5% of an adult's daily allowance of potassium, so x servings would provide x times 5%. The percentage of an adult's daily allowance of potassium, p, is 5 times the number of servings, x. Therefore, the percentage of an adult's daily allowance of potassium can be expressed as $p = 5x$.

Choices A, C, and D are incorrect and may result from incorrectly converting 5% to its decimal equivalent, which isn't necessary since p is expressed as a percentage. Additionally, choices C and D are incorrect because the context should be represented by a linear relationship, not by an exponential relationship.

## Question ID: d3b9c8d8

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard |

**Question**




The box plots summarize the masses, in kilograms, of two groups of gazelles. Based on the box plots, which of the following statements must be true?

**Answer**

A. The mean mass of group 1 is greater than the mean mass of group 2.

B. The mean mass of group 1 is less than the mean mass of group 2.

C. The median mass of group 1 is greater than the median mass of group 2.

D. The median mass of group 1 is less than the median mass of group 2.

**Correct Answer: C**

**Rationale**

Choice C is correct. The median of a data set represented in a box plot is represented by the vertical line within the box. It follows that the median mass of the gazelles in group $1$ is $25$ kilograms, and the median mass of the gazelles in group $2$ is $24$ kilograms. Since $25$ kilograms is greater than $24$ kilograms, the median mass of group $1$ is greater than the median mass of group $2$.

Choice A is incorrect. The mean mass of each of the two groups cannot be determined from the box plots.

Choice B is incorrect. The mean mass of each of the two groups cannot be determined from the box plots.

Choice D is incorrect and may result from conceptual or calculation errors.

# Question ID: 65c49824

| Assessment | Test | Domain | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Problem-Solving and Data Analysis | Percentages | Hard |

**Question**

A school district is forming a committee to discuss plans for the construction of a new high school. Of those invited to join the committee, 15% are parents of students, 45% are teachers from the current high school, 25% are school and district administrators, and the remaining 6 individuals are students. How many more teachers were invited to join the committee than school and district administrators?

**Rationale**

The correct answer is 8. The 6 students represent $(100 - 15 - 45 - 25)\% = 15\%$ of those invited to join the committee. If x people were invited to join the committee, then $0.15x = 6$. Thus, there were $\frac{6}{0.15} = 40$ people invited to join the committee. It follows that there were $0.45(40) = 18$ teachers and $0.25(40) = 10$ school and district administrators invited to join the committee. Therefore, there were 8 more teachers than school and district administrators invited to join the committee.

## Question ID: 1ea09200

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Evaluating statistical claims: Observational studies and experiments | Hard |

**Question**

A sample of 40 fourth-grade students was selected at random from a certain school. The 40 students completed a survey about the morning announcements, and 32 thought the announcements were helpful. Which of the following is the largest population to which the results of the survey can be applied?

**Answer**

A. The 40 students who were surveyed

B. All fourth-grade students at the school

C. All students at the school

D. All fourth-grade students in the county in which the school is located

**Correct Answer: B**

**Rationale**

Choice B is correct. Selecting a sample of a reasonable size at random to use for a survey allows the results from that survey to be applied to the population from which the sample was selected, but not beyond this population. In this case, the population from which the sample was selected is all fourth-grade students at a certain school. Therefore, the results of the survey can be applied to all fourth-grade students at the school.

Choice A is incorrect. The results of the survey can be applied to the 40 students who were surveyed. However, this isn't the largest group to which the results of the survey can be applied. Choices C and D are incorrect. Since the sample was selected at random from among the fourth-grade students at a certain school, the results of the survey can't be applied to other students at the school or to other fourth-grade students who weren't represented in the survey results. Students in other grades in the school or other fourth-grade students in the country may feel differently about announcements than the fourth-grade students at the school.

## Question ID: 28ff3966

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Percentages | Hard |

**Question**

A square scale drawing has a side length of $55$ inches, and $1$ inch on the drawing represents an actual distance of $19$ miles. A larger version of the same drawing is printed as a square with the side length $65\%$ longer than the side length of the previous drawing. On the larger drawing, which of the following is closest to the actual distance, in miles, represented by $1$ inch?

**Answer**

A. $6.65$

B. $11.52$

C. $31.35$

D. $35.75$

**Correct Answer: B**

**Rationale**

Choice B is correct. It's given that $1$ inch on the drawing represents an actual distance of $19$ miles and that the larger drawing has a side length that is $65\%$ longer than the side length of the previous drawing. It follows that $(1.65)(1 \text{ inch})$, or $1.65$ inches, on the larger drawing also represents an actual distance of $19$ miles. Therefore, the scale of the larger drawing can be represented as $\frac{19 \text{ miles}}{1.65 \text{ inches}}$, which is approximately $11.52$ miles per inch. Therefore, $11.52$ is closest to the actual distance, in miles, represented by $1$ inch on the larger drawing.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

## Question ID: 0ea56bb2

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Percentages | Hard |

**Question**

| Year | Subscriptions sold |
|---|---|
| 2012 | 5,600 |
| 2013 | 5,880 |

The manager of an online news service received the report above on the number of subscriptions sold by the service. The manager estimated that the percent increase from 2012 to 2013 would be double the percent increase from 2013 to 2014. How many subscriptions did the manager expect would be sold in 2014?

**Answer**

A. 6,020

B. 6,027

C. 6,440

D. 6,468

**Correct Answer: B**

**Rationale**

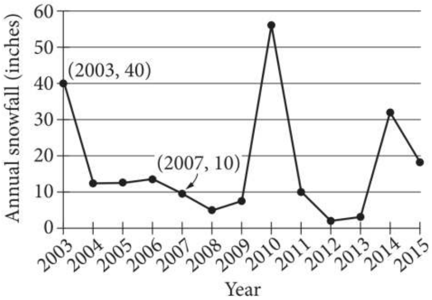
Choice B is correct. The percent increase from 2012 to 2013 was $\frac{5{,}880-5{,}600}{5{,}600}=0.05$, or 5%. Since the percent increase from 2012 to 2013 was estimated to be double the percent increase from 2013 to 2014, the percent increase from 2013 to 2014 was expected to be 2.5%. Therefore, the number of subscriptions sold in 2014 is expected to be the number of subscriptions sold in 2013 multiplied by $(1+0.025)$, or $5{,}880(1.025)=6{,}027$.

Choice A is incorrect and is the result of adding half of the value of the increase from 2012 to 2013 to the 2013 result. Choice C is incorrect and is the result adding twice the value of the increase from 2012 to 2013 to the 2013 result. Choice D is incorrect and is the result of interpreting the percent increase from 2013 to 2014 as double the percent increase from 2012 to 2013.

# Question ID: 0231050d

| Assessment | Test | Domain | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Problem-Solving and Data Analysis | Percentages | Hard |

### Question




The line graph shows the total amount of snow, in inches, recorded each year in Washington, DC, from 2003 to 2015. If $p\%$ is the percent decrease in the annual snowfall from 2003 to 2007, what is the value of p ?

### Rationale

The correct answer is 75. The percent decrease between two values is found by dividing the difference between the two values by the original value and multiplying by 100. The line graph shows that the annual snowfall in 2003 was 40 inches, and the annual snowfall in 2007 was 10 inches. Therefore, the percent decrease in the annual snowfall from 2003 to 2007 is $\left(\frac{40-10}{40}\right)(100)$, or 75. It's given that this is equivalent to $p\%$, so the value of p is 75.

## Question ID: d4413871

| Assessment | Test | Domain | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Problem-Solving and Data Analysis | Probability and conditional probability | Hard |

**Question**

| | Blood type | | | |
| --- | --- | --- | --- | --- |
| Rhesus factor | A | B | AB | O |
| + | 33 | 9 | 3 | 37 |
| – | 7 | 2 | 1 | x |

Human blood can be classified into four common blood types—A, B, AB, and O. It is also characterized by the presence $(+)$ or absence $(-)$ of the rhesus factor. The table above shows the distribution of blood type and rhesus factor for a group of people. If one of these people who is rhesus negative $(-)$ is chosen at random, the probability that the person has blood type B is $\frac{1}{9}$ . What is the value of x ?

**Rationale**

The correct answer is 8. In this group, $\frac{1}{9}$ of the people who are rhesus negative have blood type B. The total number of people who are rhesus negative in the group is $7+2+1+x$, and there are 2 people who are rhesus negative with blood type B. Therefore, $\frac{2}{(7+2+1+x)}=\frac{1}{9}$. Combining like terms on the left-hand side of the equation yields $\frac{2}{(10+x)}=\frac{1}{9}$. Multiplying both sides of this equation by 9 yields $\frac{18}{(10+x)}=1$, and multiplying both sides of this equation by $(10+x)$ yields $18=10+x$. Subtracting 10 from both sides of this equation yields $8=x$.

## Question ID: f6c18a66

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard |

**Question**

For data set A, the table summarizes the distribution of the number of deliveries received by an office each day during a period of $11$ days.

| Deliveries | Days |
|---|---|
| $0$ | $2$ |
| $3$ | $2$ |
| $4$ | $2$ |
| $5$ | $2$ |
| $6$ | $1$ |
| $7$ | $1$ |
| $13$ | $1$ |

The data value $13$ was recorded in error and is removed from data set A to create data set B, which consists of the remaining $10$ data values. Which statement best compares the median of data set A and the median of data set B?

**Answer**

A. The median of data set B is less than the median of data set A.

B. The median of data set B is greater than the median of data set A.

C. The median of data set B is equal to the median of data set A.

D. There is not enough information to compare the medians of the two data sets.

**Correct Answer: C**

**Rationale**

Choice C is correct. It's given that for data set A, the table summarizes the number of deliveries received by an office each day over a period of $11$ days. If a data set contains an odd number of data values, the median is represented by the middle data value in the list when the data values are listed in ascending or descending order. Since data set A contains $11$ data values in ascending order, the median of data set A is the $6\text{th}$ value. Based on the number of days for each value shown in the table, the $6\text{th}$ value is $4$, so the median of data set A is $4$. It's also given that the data value $13$ was recorded in error and is removed from data set A to create data set B. It follows that data set B contains $10$ data values in ascending order. If a data set contains an even number of data values, the median is between the two middle data values when the values are listed in ascending or descending order. It follows that the median of data set B is between the $5\text{th}$ and $6\text{th}$ values. Based on the number of days for each value shown in the table, the $5\text{th}$ and $6\text{th}$ values are both $4$, so the median of data set B is $4$. Therefore, the median of data set B is equal to the median of data set A.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

## Question ID: 190be2fc

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard |

**Question**

Data set A consists of $10$ positive integers less than $60$. The list shown gives $9$ of the integers from data set A.

$$43, 45, 44, 43, 38, 39, 40, 46, 40$$

The mean of these $9$ integers is $42$. If the mean of data set A is an integer that is greater than $42$, what is the value of the largest integer from data set A?

**Correct Answer: 52**

**Rationale**

The correct answer is $52$. The mean of a data set is calculated by dividing the sum of the data values by the number of values. It's given that data set A consists of $10$ values, $9$ of which are shown. Let $x$ represent the $10\text{th}$ data value in data set A, which isn't shown. The mean of data set A can be found using the expression $\frac{43+45+44+43+38+39+40+46+40+x}{10}$, or $\frac{378+x}{10}$. It's given that the mean of the $9$ values shown is $42$ and that the mean of all $10$ numbers is greater than $42$. Consequently, the $10\text{th}$ data value, $x$, is larger than $42$. It's also given that the data values in data set A are positive integers less than $60$. Thus, $42 < x < 60$. Finally, it's given that the mean of data set A is an integer. This means that the sum of the $10$ data values, $378 + x$, is divisible by $10$. Thus, $378 + x$ must have a ones digit of $0$. It follows that $x$ must have a ones digit of $2$. Since $42 < x < 60$ and $x$ has a ones digit of $2$, the only possible value of $x$ is $52$. Since $52$ is larger than any of the integers shown, the largest integer from data set A is $52$.

# Question ID: db88a933

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard |

**Question**

Data Set Q

| Value | Frequency |
|---|---|
| $a$ | $110$ |
| $b$ | $30$ |
| $c$ | $0$ |

Data Set R

| Value | Frequency |
|---|---|
| $a$ | $0$ |
| $b$ | $3$ |
| $c$ | $11$ |

Data sets Q and R are summarized in the tables shown, where $a$, $b$, and $c$ are positive, consecutive integers and $a < b < c$. Which statement comparing the means of the data sets is true?

**Answer**

A. The mean of data set R is $\frac{2(11)}{14}$ less than the mean of data set Q.

B. The mean of data set R is $\frac{2(11)}{14}$ greater than the mean of data set Q.

C. The mean of data set Q is $10$ times the mean of data set R.

D. The means of the two data sets are equal.

**Correct Answer: B**

**Rationale**

Choice B is correct. The mean of a data set is the sum of the values of the data set divided by the total number of values in the data set. When a data set is represented in a frequency table, the sum of the values in the data set is the sum of the products of each value and its frequency. For data set Q, the sum of the products of each value and its frequency is $110a + 30b + 0c$, or $110a + 30b$, and the number of values is $110 + 30 + 0$, or $140$. Therefore, the mean of data set Q is $\frac{110a+30b}{140}$. For data set R, the sum of the products of each value and its frequency is $0a + 3b + 11c$, or $3b + 11c$, and the number of values is $0 + 3 + 11$, or $14$. Therefore, the mean of data set R is $\frac{3b+11c}{14}$. Since $a$, $b$, and $c$ are positive, consecutive integers and $a < b < c$, it follows that $b = a + 1$ and $c = a + 2$. Substituting $a + 1$ for $b$ in the mean of data set Q yields $\frac{110a+30(a+1)}{140}$, or $\frac{110a+30a+30}{140}$, which is equivalent to $\frac{140a+30}{140}$, or $a + \frac{3}{14}$. Substituting $a + 1$ for $b$ and $a + 2$ for $c$ in the mean of data set R yields $\frac{3(a+1)+11(a+2)}{14}$, or $\frac{3a+3+11a+2(11)}{14}$, which is equivalent to $\frac{14a+3+2(11)}{14}$, or $a + \frac{3}{14} + \frac{2(11)}{14}$. Subtracting the mean of data set Q from the mean of data set R yields $\left(a + \frac{3}{14} + \frac{2(11)}{14}\right) - \left(a + \frac{3}{14}\right)$, or $\frac{2(11)}{14}$. Therefore, the mean of data set R is $\frac{2(11)}{14}$ greater than the mean of data set Q.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

## Question ID: c178d4da

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard |

**Question**

| Value | Data set A frequency | Data set B frequency |
|---|---|---|
| $30$ | $2$ | $9$ |
| $34$ | $4$ | $7$ |
| $38$ | $5$ | $5$ |
| $42$ | $7$ | $4$ |
| $46$ | $9$ | $2$ |

Data set A and data set B each consist of $27$ values. The table shows the frequencies of the values for each data set. Which of the following statements best compares the means of the two data sets?

**Answer**

A. The mean of data set A is greater than the mean of data set B.

B. The mean of data set A is less than the mean of data set B.

C. The mean of data set A is equal to the mean of data set B.

D. There is not enough information to compare the means of the data sets.

**Correct Answer: A**

**Rationale**

Choice A is correct. The mean value of a data set is the sum of the values of the data set divided by the number of values in the data set. When a data set is represented in a frequency table, the sum of the values in the data set is the sum of the products of each value and its frequency. For data set A, the sum of products of each value and its frequency is $30(2) + 34(4) + 38(5) + 42(7) + 46(9)$, or $1,094$. It's given that there are $27$ values in data set A. Therefore, the mean of data set A is $\frac{1,094}{27}$, or approximately $40.52$. Similarly, the mean of data B is $\frac{958}{27}$, or approximately $35.48$. Therefore, the mean of data set A is greater than the mean of data set B.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

## Question ID: 457d2f2c

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard |

**Question**

A data set of 27 different numbers has a mean of 33 and a median of 33. A new data set is created by adding 7 to each number in the original data set that is greater than the median and subtracting 7 from each number in the original data set that is less than the median. Which of the following measures does NOT have the same value in both the original and new data sets?

**Answer**

A. Median

B. Mean

C. Sum of the numbers

D. Standard deviation

**Correct Answer: D**

**Rationale**

Choice D is correct. When a data set has an odd number of elements, the median can be found by ordering the values from least to greatest and determining the middle value. Out of the 27 different numbers in this data set, 13 numbers are below the median, one number is exactly 33, and 13 numbers are above the median. When 7 is subtracted from each number below the median and added to each number above the median, the data spread out from the median. Since the median of this data set, 33, is equivalent to the mean of the data set, the data also spread out from the mean. Since standard deviation is a measure of how spread out the data are from the mean, a greater spread from the mean indicates an increased standard deviation.

Choice A is incorrect. All the numbers less than the median decrease and all the numbers greater than the median increase, but the median itself doesn't change. Choices B and C are incorrect. The mean of a data set is found by dividing the sum of the values by the number of values. The net change from subtracting 7 from 13 numbers and adding 7 to 13 numbers is zero. Therefore, neither the mean nor the sum of the numbers changes.

# Question ID: d6af3572

| Assessment | Test | Domain | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Problem-Solving and Data Analysis | Two-variable data: Models and scatterplots | Hard |

**Question**


The scatterplot above shows the federal-mandated minimum wage every 10 years between 1940 and 2010. A line of best fit is shown, and its equation is $y = 0.096x - 0.488$. What does the line of best fit predict about the increase in the minimum wage over the 70-year period?

**Answer**

A. Each year between 1940 and 2010, the average increase in minimum wage was 0.096 dollars.

B. Each year between 1940 and 2010, the average increase in minimum wage was 0.49 dollars.

C. Every 10 years between 1940 and 2010, the average increase in minimum wage was 0.096 dollars.

D. Every 10 years between 1940 and 2010, the average increase in minimum wage was 0.488 dollars.

**Correct Answer: A**

**Rationale**

Choice A is correct. The given equation is in slope-intercept form, or $y = mx + b$, where m is the value of the slope of the line of best fit. Therefore, the slope of the line of best fit is 0.096. From the definition of slope, it follows that an increase of 1 in the x-value corresponds to an increase of 0.096 in the y-value. Therefore, the line of best fit predicts that for each year between 1940 and 2010, the minimum wage will increase by 0.096 dollar per hour.

Choice B is incorrect and may result from using the y-coordinate of the y-intercept as the average increase, instead of the slope. Choice C is incorrect and may result from using the 10-year increments given on the x-axis to incorrectly interpret the slope of the line of best fit. Choice D is incorrect and may result from using the y-coordinate of the y-intercept as the average increase, instead of the slope, and from using the 10-year increments given on the x-axis to incorrectly interpret the slope of the line of best fit.

## Question ID: 89ff6a0a

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Probability and conditional probability | Hard |

**Question**

For $100$ buttons, the table summarizes the distribution by group and diameter.

| Group | Diameter (millimeters) | | |
|---|---|---|---|
| | **Less than $20$** | **$20$ to $30$** | **Greater than $30$** |
| **Group 1** | $12$ | $8$ | $3$ |
| **Group 2** | $0$ | $17$ | $18$ |
| **Group 3** | $10$ | $32$ | $0$ |

One of these buttons will be selected at random. What is the probability of selecting a button with a diameter that is less than or equal to $30$ millimeters, given that it is **not** in group 2? (Express your answer as a decimal or fraction, not as a percent.)

**Correct Answer: .9538, 62/65**

**Rationale**

The correct answer is $\frac{62}{65}$. It's given that one button will be selected at random out of a total of $100$ buttons and that the table shows the distribution of buttons by group and diameter. The probability of selecting a button with a diameter less than or equal to $30$ millimeters, given that it is not in group 2, is the number of buttons with a diameter less than or equal to $30$ millimeters that are not in group 2 divided by the total number of buttons not in group 2. Based on the table, the buttons not in group 2 are those in group 1 and group 3. In group 1, there are $12$ buttons with a diameter less than $20$ millimeters and $8$ buttons with a diameter from $20$ to $30$ millimeters, which yields $12 + 8$, or $20$, buttons with a diameter less than or equal to $30$ millimeters. In group 3, there are $10$ buttons with a diameter less than $20$ millimeters and $32$ buttons with a diameter from $20$ to $30$ millimeters, which yields $10 + 32$, or $42$, buttons with a diameter less than or equal to $30$ millimeters. Therefore, the total number of buttons with a diameter less than or equal to $30$ millimeters that are not in group 2 is $20 + 42$, or $62$. The total number of buttons that are not in group 2 is the sum of all buttons in group 1 and group 3. There are $12 + 8 + 3$, or $23$, buttons in group 1 and $10 + 32 + 0$, or $42$, buttons in group 3. Therefore, the total number of buttons in groups 1 and 3 is $23 + 42$, or $65$. It follows that the probability of selecting a button with a diameter less than or equal to $30$ millimeters, given that it is not in group 2, is $\frac{62}{65}$. Note that 62/65 or .9538 are examples of ways to enter a correct answer.

# Question ID: 8bb94668

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Percentages | Hard |

**Question**

The speeds of objects A, B, and C are $a$ meters per second, $b$ meters per second, and $c$ meters per second, respectively. If the speed of object A is $5{,}900\%$ of the speed of object C and the speed of object C is $0.008\%$ of the speed of object B, which expression represents the value of $a + b$ in terms of $c$?

**Answer**

A. $1{,}840c$

B. $5{,}908c$

C. $6{,}025c$

D. $12{,}559c$

**Correct Answer: D**

**Rationale**

Choice D is correct. It's given that the speeds of objects A, B, and C are $a$ meters per second, $b$ meters per second, and $c$ meters per second, respectively. If the speed of object A is $5{,}900\%$ of the speed of object C, this means that $a = \frac{5{,}900}{100}c$, or $a = 59c$. If the speed of object C is $0.008\%$ of the speed of object B, this means that $c = \frac{0.008}{100}b$, or $c = 0.00008b$. Dividing both sides of this equation by $0.00008$ yields $\frac{1}{0.00008}c = b$, or $12{,}500c = b$. Since $a = 59c$ and $b = 12{,}500c$, it follows that $a + b = 59c + 12{,}500c$, or $a + b = 12{,}559c$. Therefore, the expression $12{,}559c$ represents the value of $a + b$ in terms of $c$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

## Question ID: 3638f413

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Ratios, rates, proportional relationships, and units | Hard |

**Question**

Jeremy deposited x dollars in his investment account on January 1, 2001. The amount of money in the account doubled each year until Jeremy had 480 dollars in his investment account on January 1, 2005. What is the value of x ?

**Rationale**

The correct answer is 30. The situation can be represented by the equation $x(2^4) = 480$, where the 2 represents the fact that the amount of money in the account doubled each year and the 4 represents the fact that there are 4 years between January 1, 2001, and January 1, 2005. Simplifying $x(2^4) = 480$ gives $16x = 480$. Therefore, $x = 30$.

## Question ID: 1142af44

| Assessment | Test | Domain | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard |

**Question**

| Value | Frequency |
| --- | --- |
| 1 | a |
| 2 | 2a |
| 3 | 3a |
| 4 | 2a |
| 5 | a |

The frequency distribution above summarizes a set of data, where *a* is a positive integer. How much greater is the mean of the set of data than the median?

**Answer**

A. 0

B. 1

C. 2

D. 3

**Correct Answer: A**

**Rationale**

Choice A is correct. Since the frequencies of values less than the middle value, 3, are the same as the frequencies of the values greater than 3, the set of data has a symmetric distribution. When a set of data has a symmetric distribution, the mean and median values are equal. Therefore, the mean is 0 greater than the median.

Choices B, C, and D are incorrect and may result from misinterpreting the set of data.

## Question ID: 1e8ccffd

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard |

**Question**

The mean score of 8 players in a basketball game was 14.5 points. If the highest individual score is removed, the mean score of the remaining 7 players becomes 12 points. What was the highest score?

**Answer**

A. 20

B. 24

C. 32

D. 36

**Correct Answer: C**

**Rationale**

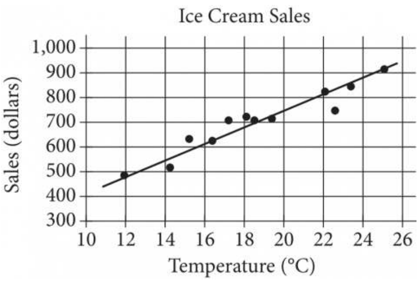
Choice C is correct. If the mean score of 8 players is 14.5, then the total of all 8 scores is $14.5 \times 8 = 116$. If the mean of 7 scores is 12, then the total of all 7 scores is $12 \times 7 = 84$. Since the set of 7 scores was made by removing the highest score of the set of 8 scores, then the difference between the total of all 8 scores and the total of all 7 scores is equal to the removed score: $116 - 84 = 32$.

Choice A is incorrect because if 20 is removed from the group of 8 scores, then the mean score of the remaining 7 players is $\frac{(14.5 \times 8) - 20}{7}$ is approximately 13.71, not 12. Choice B is incorrect because if 24 is removed from the group of 8 scores, then the mean score of the remaining 7 players is $\frac{(14.5 \times 8) - 24}{7}$ is approximately 13.14, not 12. Choice D is incorrect because if 36 is removed from the group of 8 scores, then the mean score of the remaining 7 players is $\frac{(14.5 \times 8) - 36}{7}$ or approximately 11.43, not 12.

# Question ID: 1e1027a7

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Two-variable data: Models and scatterplots | Hard |

**Question**




The scatterplot above shows a company's ice cream sales d, in dollars, and the high temperature t, in degrees Celsius (°C), on 12 different days. A line of best fit for the data is also shown. Which of the following could be an equation of the line of best fit?

**Answer**

A. $d = 0.03t + 402$

B. $d = 10t + 402$

C. $d = 33t + 300$

D. $d = 33t + 84$

**Correct Answer: D**

**Rationale**

Choice D is correct. On the line of best fit, d increases from approximately 480 to 880 between $t = 12$ and $t = 24$. The slope of the line of best fit is the difference in d-values divided by the difference in t-values, which gives $\frac{880 - 480}{24 - 12} = \frac{400}{12}$, or approximately 33. Writing the equation of the line of best fit in slope-intercept form gives $d = 33t + b$, where b is the y-coordinate of the y-intercept. This equation is satisfied by all points on the line, so $d = 480$ when $t = 12$. Thus, $480 = 33(12) + b$, which is equivalent to $480 = 396 + b$. Subtracting 396 from both sides of this equation gives $b = 84$. Therefore, an equation for the line of best fit could be $d = 33t + 84$.

Choice A is incorrect and may result from an error in calculating the slope and misidentifying the y-coordinate of the y-intercept of the graph as the value of d at $t = 10$ rather than the value of d at $t = 0$. Choice B is incorrect and may result from using the smallest value of t on the graph as the slope and misidentifying the y-coordinate of the y-intercept of the graph as the value of d at $t = 10$ rather than the value of d at $t = 0$. Choice C is incorrect and may result from misidentifying the y-coordinate of the y-intercept as the smallest value of d on the graph.

## Question ID: 8637294f

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Ratios, rates, proportional relationships, and units | Hard |

**Question**

If $\frac{4a}{b} = 6.7$ and $\frac{a}{bn} = 26.8$, what is the value of $n$?

**Correct Answer: .0625, 1/16**

**Rationale**

The correct answer is $.0625$. It's given that $\frac{4a}{b} = 6.7$ and $\frac{a}{bn} = 26.8$. The equation $\frac{4a}{b} = 6.7$ can be rewritten as $(4)\left(\frac{a}{b}\right) = 6.7$. Dividing both sides of this equation by $4$ yields $\frac{a}{b} = 1.675$. The equation $\frac{a}{bn} = 26.8$ can be rewritten as $\left(\frac{a}{b}\right)\left(\frac{1}{n}\right) = 26.8$. Substituting $1.675$ for $\frac{a}{b}$ in this equation yields $(1.675)\left(\frac{1}{n}\right) = 26.8$, or $\frac{1.675}{n} = 26.8$. Multiplying both sides of this equation by $n$ yields $1.675 = 26.8n$. Dividing both sides of this equation by $26.8$ yields $n = 0.0625$. Therefore, the value of $n$ is $0.0625$. Note that .0625, 0.062, 0.063, and 1/16 are examples of ways to enter a correct answer.

## Question ID: 308084c5

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Inference from sample statistics and margin of error | Hard |

**Question**

| Sample | Percent in favor | Margin of error |
|---|---|---|
| A | 52% | 4.2% |
| B | 48% | 1.6% |

The results of two random samples of votes for a proposition are shown above. The samples were selected from the same population, and the margins of error were calculated using the same method. Which of the following is the most appropriate reason that the margin of error for sample A is greater than the margin of error for sample B?

**Answer**

A. Sample A had a smaller number of votes that could not be recorded.

B. Sample A had a higher percent of favorable responses.

C. Sample A had a larger sample size.

D. Sample A had a smaller sample size.

**Correct Answer: D**

**Rationale**

Choice D is correct. Sample size is an appropriate reason for the margin of error to change. In general, a smaller sample size increases the margin of error because the sample may be less representative of the whole population.

Choice A is incorrect. The margin of error will depend on the size of the sample of recorded votes, not the number of votes that could not be recorded. In any case, the smaller number of votes that could not be recorded for sample A would tend to decrease, not increase, the comparative size of the margin of error. Choice B is incorrect. Since the percent in favor for sample A is the same distance from 50% as the percent in favor for sample B, the percent of favorable responses doesn't affect the comparative size of the margin of error for the two samples. Choice C is incorrect. If sample A had a larger margin of error than sample B, then sample A would tend to be less representative of the population. Therefore, sample A is not likely to have a larger sample size.

## Question ID: 7d721177

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Ratios, rates, proportional relationships, and units | Hard |

**Question**

The density of a certain type of wood is $353$ kilograms per cubic meter. A sample of this type of wood is in the shape of a cube and has a mass of $345$ kilograms. To the nearest hundredth of a meter, what is the length of one edge of this sample?

**Answer**

A. $0.98$

B. $0.99$

C. $1.01$

D. $1.02$

**Correct Answer: B**

**Rationale**

Choice B is correct. It's given that the density of a certain type of wood is $353$ kilograms per cubic meter $\left(\text{kg/m}^3\right)$, and a sample of this type of wood has a mass of $345\ \text{kg}$. Let $x$ represent the volume, in $\text{m}^3$, of the sample. It follows that the relationship between the density, mass, and volume of this sample can be written as $\frac{353\ \text{kg}}{1\ \text{m}^3} = \frac{345\ \text{kg}}{x\ \text{m}^3}$, or $353 = \frac{345}{x}$. Multiplying both sides of this equation by $x$ yields $353x = 345$. Dividing both sides of this equation by $353$ yields $x = \frac{345}{353}$. Therefore, the volume of this sample is $\frac{345}{353}\ \text{m}^3$. Since it's given that the sample of this type of wood is a cube, it follows that the length of one edge of this sample can be found using the volume formula for a cube, $V = s^3$, where $V$ represents the volume, in $\text{m}^3$, and $s$ represents the length, in m, of one edge of the cube. Substituting $\frac{345}{353}$ for $V$ in this formula yields $\frac{345}{353} = s^3$. Taking the cube root of both sides of this equation yields $\sqrt[3]{\frac{345}{353}} = s$, or $s \approx 0.99$. Therefore, the length of one edge of this sample to the nearest hundredth of a meter is $0.99$.

Choices A, C, and D are incorrect and may result from conceptual or calculation errors.

## Question ID: 67c0200a

| Assessment | Test | Math | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Problem-Solving and Data Analysis | Percentages | Hard |

**Question**

The number $a$ is $70\%$ less than the positive number $b$. The number $c$ is $80\%$ greater than $a$. The number $c$ is how many times $b$?

**Correct Answer: .54, 27/50**

**Rationale**

The correct answer is $.54$. It's given that the number $a$ is $70\%$ less than the positive number $b$. Therefore, $a = \left(1 - \frac{70}{100}\right)b$, which is equivalent to $a = (1 - 0.70)b$, or $a = 0.30b$. It's also given that the number $c$ is $80\%$ greater than $a$. Therefore, $c = \left(1 + \frac{80}{100}\right)a$, which is equivalent to $c = (1 + 0.80)a$, or $c = 1.80a$. Since $a = 0.30b$, substituting $0.30b$ for $a$ in the equation $c = 1.80a$ yields $c = 1.80(0.30b)$, or $c = 0.54b$. Thus, $c$ is $0.54$ times $b$. Note that .54 and 27/50 are examples of ways to enter a correct answer.

## Question ID: 7d68096f

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Evaluating statistical claims: Observational studies and experiments | Hard |

**Question**

A trivia tournament organizer wanted to study the relationship between the number of points a team scores in a trivia round and the number of hours that a team practices each week. For the study, the organizer selected $55$ teams at random from all trivia teams in a certain tournament. The table displays the information for the $40$ teams in the sample that practiced for at least $3$ hours per week.

| Hours practiced | Number of points per round | | |
|---|---|---|---|
| | 6 to 13 points | 14 or more points | Total |
| 3 to 5 hours | $6$ | $4$ | $10$ |
| More than 5 hours | $4$ | $26$ | $30$ |
| Total | $10$ | $30$ | $40$ |

Which of the following is the largest population to which the results of the study can be generalized?

**Answer**

A. All trivia teams in the tournament that scored $14$ or more points in the round

B. The $55$ trivia teams in the sample

C. The $40$ trivia teams in the sample that practiced for at least $3$ hours per week

D. All trivia teams in the tournament

**Correct Answer: D**

**Rationale**

Choice D is correct. It's given that the organizer selected $55$ teams at random from all trivia teams in the tournament. A table is also given displaying the information for the $40$ teams in the sample that practiced for at least $3$ hours per week. Selecting a sample of a reasonable size at random to use for a survey allows the results from that survey to be applied to the population from which the sample was selected, but not beyond this population. Thus, only the sampling method information is necessary to determine the largest population to which the results of the study can be generalized. Since the organizer selected the sample at random from all trivia teams in the tournament, the largest population to which the results of the study can be generalized is all trivia teams in the tournament.

Choice A is incorrect. The sample was selected at random from all trivia teams in the tournament, not just from the teams that scored an average of $14$ or more points per round.

Choice B is incorrect. If a study uses a sample selected at random from a population, the results of the study can be generalized to the population, not just the sample.

Choice C is incorrect. If a study uses a sample selected at random from a population, the results of the study can be generalized to the population, not just a subset of the sample.

## Question ID: 585de39a

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Probability and conditional probability | Hard |

**Question**

On May 10, 2015, there were 83 million Internet subscribers in Nigeria. The major Internet providers were MTN, Globacom, Airtel, Etisalat, and Visafone. By September 30, 2015, the number of Internet subscribers in Nigeria had increased to 97 million. If an Internet subscriber in Nigeria on September 30, 2015, is selected at random, the probability that the person selected was an MTN subscriber is 0.43. There were p million MTN subscribers in Nigeria on September 30, 2015. To the nearest integer, what is the value of p ?

**Rationale**

The correct answer is 42. It's given that in Nigeria on September 30, 2015, the probability of selecting an MTN subscriber from all Internet subscribers is 0.43, that there were p million, or $p(1{,}000{,}000)$, MTN subscribers, and that there were 97 million, or 97,000,000, Internet subscribers. The probability of selecting an MTN subscriber from all Internet subscribers can be found by dividing the number of MTN subscribers by the total number of Internet subscribers. Therefore, the equation $\frac{p(1{,}000{,}000)}{97{,}000{,}000} = 0.43$ can be used to solve for p. Dividing 1,000,000 from the numerator and denominator of the expression on the left-hand side yields $\frac{p}{97} = 0.43$. Multiplying both sides of this equation by 97 yields $p = (0.43)(97) = 41.71$, which, to the nearest integer, is 42.

## Question ID: 4ff597db

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard |

**Question**

The mean amount of time that the 20 employees of a construction company have worked for the company is 6.7 years. After one of the employees leaves the company, the mean amount of time that the remaining employees have worked for the company is reduced to 6.25 years. How many years did the employee who left the company work for the company?

**Answer**

A. 0.45

B. 2.30

C. 9.00

D. 15.25

**Correct Answer: D**

**Rationale**

Choice D is correct. The mean amount of time that the 20 employees worked for the company is 6.7 years. This means that the total number of years all 20 employees worked for the company is (6.7)(20) = 134 years. After the employee left, the mean amount of time that the remaining 19 employees worked for the company is 6.25 years. Therefore, the total number of years all 19 employees worked for the company is (6.25)(19) = 118.75 years. It follows that the number of years that the employee who left had worked for the company is 134 – 118.75 = 15.25 years.

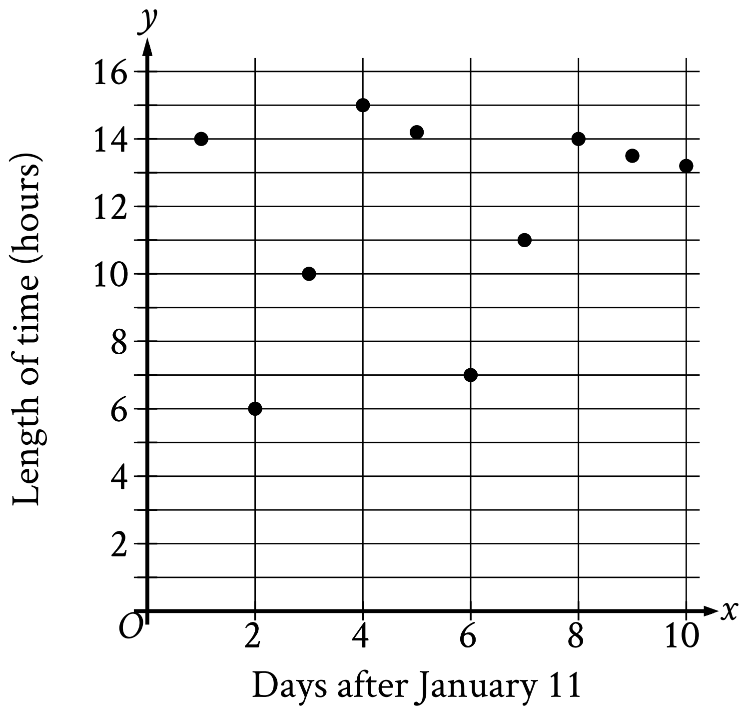
Choice A is incorrect; this is the change in the mean, which isn't the same as the amount of time worked by the employee who left. Choice B is incorrect and likely results from making the assumption that there were still 20 employees, rather than 19, at the company after the employee left and then subtracting the original mean of 6.7 from that result. Choice C is incorrect and likely results from making the assumption that there were still 20 employees, rather than 19, at the company after the employee left.

## Question ID: 7b52985c

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Two-variable data: Models and scatterplots | Hard |

**Question**

The scatterplot shows the relationship between the length of time $y$, in hours, a certain bird spent in flight and the number of days after January $11$, $x$.




What is the average rate of change, in hours per day, of the length of time the bird spent in flight on January $13$ to the length of time the bird spent in flight on January $15$?

**Correct Answer: 4.5, 9/2**

**Rationale**

The correct answer is $\frac{9}{2}$. It's given that the scatterplot shows the relationship between the length of time $y$, in hours, a certain bird spent in flight and the number of days after January $11$, $x$. Since January $13$ is $2$ days after January $11$, it follows that January $13$ corresponds to an *x*-value of $2$ in the scatterplot. In the scatterplot, when $x = 2$, the corresponding value of $y$ is $6$. In other words, on January $13$, the bird spent $6$ hours in flight. Since January $15$ is $4$ days after January $11$, it follows that January $15$ corresponds to an *x*-value of $4$ in the scatterplot. In the scatterplot, when $x = 4$, the corresponding value of $y$ is $15$. In other words, on January $15$, the bird spent $15$ hours in flight. Therefore, the average rate of change, in hours per day, of the length of time the bird spent in flight on January $13$ to the length of time the bird spent in flight on January $15$ is the difference in the length of time, in hours, the bird spent in flight divided by the difference in the number of days after January $11$, or $\frac{15-6}{4-2}$, which is equivalent to $\frac{9}{2}$. Note that 9/2 and 4.5 are examples of ways to enter a correct answer.

## Question ID: 7ce2830a

| Assessment | Test | Domain | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Problem-Solving and Data Analysis | Evaluating statistical claims: Observational studies and experiments | Hard |

**Question**

A psychologist designed and conducted a study to determine whether playing a certain educational game increases middle school students' accuracy in adding fractions. For the study, the psychologist chose a random sample of 35 students from all of the students at one of the middle schools in a large city. The psychologist found that students who played the game showed significant improvement in accuracy when adding fractions. What is the largest group to which the results of the study can be generalized?

**Answer**

A. The 35 students in the sample

B. All students at the school

C. All middle school students in the city

D. All students in the city

**Correct Answer: B**

**Rationale**

Choice B is correct. The largest group to which the results of a study can be generalized is the population from which the random sample was chosen. In this case, the psychologist chose a random sample from all students at one particular middle school. Therefore, the largest group to which the results can be generalized is all the students at the school.

Choice A is incorrect because this isn't the largest group the results can be generalized to. Choices C and D are incorrect because these groups are larger than the population from which the random sample was chosen. Therefore, the sample isn't representative of these groups.

# Question ID: 6a715bed

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Probability and conditional probability | Hard |

**Question**

The table summarizes the distribution of age and assigned group for $90$ participants in a study.

| | $0–9$ years | $10–19$ years | $20+$ years | Total |
|---|---|---|---|---|
| **Group A** | $7$ | $14$ | $9$ | $30$ |
| **Group B** | $6$ | $4$ | $20$ | $30$ |
| **Group C** | $17$ | $12$ | $1$ | $30$ |
| **Total** | $30$ | $30$ | $30$ | $90$ |

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least $10$ years of age? (Express your answer as a decimal or fraction, not as a percent.)

**Correct Answer: .3833, 23/60**

**Rationale**

The correct answer is $\frac{23}{60}$. It's given that one of the participants will be selected at random. The probability of selecting a participant from group A given that the participant is at least $10$ years of age is the number of participants in group A who are at least $10$ years of age divided by the total number of participants who are at least $10$ years of age. The table shows that in group A, there are $14$ participants who are $10–19$ years of age and $9$ participants who are $20+$ years of age. Therefore, there are $14 + 9$, or $23$, participants in group A who are at least $10$ years of age. The table also shows that there are a total of $30$ participants who are $10–19$ years of age and $30$ participants who are $20+$ years of age. Therefore, there are a total of $30 + 30$, or $60$, participants who are at least $10$ years of age. It follows that the probability of selecting a participant from group A given that the participant is at least $10$ years of age is $\frac{23}{60}$. Note that 23/60, .3833, and 0.383 are examples of ways to enter a correct answer.

# Question ID: ed6b7e5f

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Probability and conditional probability | Hard |

**Question**

The table gives the distribution of topping and type of crust for customer orders at a pizza place.

| | Type of crust | |
|---|---|---|
| **Topping** | **Thin** | **Thick** |
| **Extra cheese** | $60$ | $30$ |
| **Pepperoni** | $20$ | $30$ |
| **Tomato** | $80$ | $20$ |

If one of these customer orders is selected at random, what is the probability of selecting an order with thin crust, given the topping is pepperoni? (Express your answer as a decimal or fraction, not as a percent.)

**Correct Answer: 0.4, 2/5**

**Rationale**

The correct answer is $\frac{2}{5}$. Since the order will be selected at random, the probability of selecting an order with thin crust, given the topping is pepperoni, is equal to the number of orders with thin crust where the topping is pepperoni divided by the total number of orders where the topping is pepperoni. According to the table, there are $20$ orders with thin crust where the topping is pepperoni. According to the table, there are $30$ orders with thick crust where the topping is pepperoni. Therefore, there are a total of $20 + 30$, or $50$, orders where the topping is pepperoni. Thus, the probability of selecting an order with thin crust, given the topping is pepperoni, is $\frac{20}{50}$, or $\frac{2}{5}$. Note that 2/5 and .4 are examples of ways to enter a correct answer.

# Question ID: 4c5ea142

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Ratios, rates, proportional relationships, and units | Hard |

**Question**

On average, a certain plant grows $87$ millimeters every $m$ months. At this rate, which expression represents the number of millimeters, on average, the plant grows every $k$ years?

**Answer**

A. $\frac{29m}{4k}$

B. $\frac{29k}{4m}$

C. $\frac{1{,}044m}{k}$

D. $\frac{1{,}044k}{m}$

**Correct Answer: D**

**Rationale**

Choice D is correct. It's given that on average, a certain plant grows $87$ millimeters every $m$ months. This average rate can be written as $\frac{87}{m}$ millimeters per month. Since there are $12$ months in $1$ year, it follows that there are $12k$ months in $k$ years. Therefore, the number of millimeters, on average, the plant grows every $k$ years can be calculated by multiplying the rate $\frac{87}{m}$ millimeters per month by $12k$ months, which yields $\left(\frac{87}{m}\right)(12k)$, or $\frac{1{,}044k}{m}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

## Question ID: 61f61789

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Ratios, rates, proportional relationships, and units | Hard |

**Question**

To study the moisture content in a group of trees, samples from the trunk of each tree were taken from $25$ trees and cut in the shape of a cube. The length of the edge of one of these cubes is $2.00$ centimeters. If this cube has a mass of $2.56$ grams, what is the density of this cube, in grams per cubic centimeter?

**Correct Answer: 0.32, 8/25**

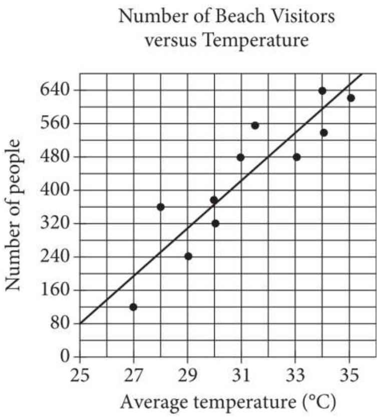
**Rationale**

The correct answer is $.32$. The volume of a cube is given by the formula $V = s^3$, where $s$ is the length of an edge of the cube. It's given that each edge of the cube has a length of $2.00$ centimeters. Substituting $2.00$ centimeters for $s$ in the formula $V = s^3$ yields $V = (2.00 \text{ centimeters})^3$, or $V = 8.00$ cubic centimeters. It's given that the cube has a mass of $2.56$ grams. Dividing the mass, in grams, of the cube by the volume, in cubic centimeters, of the cube gives its density, in grams per cubic centimeters. Therefore, the density of the cube is $\frac{2.56 \text{ grams}}{8.00 \text{ cubic centimeters}}$, or $.32$ grams per cubic centimeter. Note that .32 and 8/25 are examples of ways to enter a correct answer.

## Question ID: d0430601

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Two-variable data: Models and scatterplots | Hard |

**Question**




Each dot in the scatterplot above represents the temperature and the number of people who visited a beach in Lagos, Nigeria, on one of eleven different days. The line of best fit for the data is also shown. The line of best fit for the data has a slope of approximately 57. According to this estimate, how many additional people per day are predicted to visit the beach for each 5°C increase in average temperature?

**Rationale**

The correct answer is 285. The number of people predicted to visit the beach each day is represented by the y-values of the line of best fit, and the average temperature, in degrees Celsius ($^\circ C$), is represented by the x-values. Since the slope of the line of best fit is approximately 57, the y-value, or the number of people predicted to visit the beach each day, increases by 57 for every x-value increase of 1, or every $1^\circ C$ increase in average temperature. Therefore, an increase of $5^\circ C$ in average temperature corresponds to a y-value increase of $57(5) = 285$ additional people per day predicted to visit the beach.

## Question ID: 1b96c116

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Ratios, rates, proportional relationships, and units | Hard |

**Question**

A rectangle has an area of $3{,}168$ square inches. What is the area, in **square feet**, of this rectangle? $(1 \text{ foot} = 12 \text{ inches})$

**Answer**

A. $22$

B. $56.3$

C. $264$

D. $675.4$

**Correct Answer: A**

**Rationale**

Choice A is correct. It's given that $1 \text{ foot} = 12 \text{ inches}$. It follows that $1^2 \text{ square foot} = 12^2 \text{ square inches}$, or $1 \text{ square foot} = 144 \text{ square inches}$. Thus, $3{,}168$ square inches is equivalent to $\left(3{,}168 \text{ square inches}\right)\left(\frac{1 \text{ square foot}}{144 \text{ square inches}}\right)$, or $22$ square feet.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $\frac{3{,}168}{12}$, not $\frac{3{,}168}{144}$.

Choice D is incorrect and may result from conceptual or calculation errors.

## Question ID: b695711c

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Evaluating statistical claims: Observational studies and experiments | Hard |

**Question**

A statistician conducted a study to investigate the relationship between the average number of minutes piano students practiced each week and the percentage of correct notes the students played during a recital. For the study, the statistician selected a sample of $85$ piano students at random from all the piano students in a certain region who were in their second year with their conservatory. The table shows the information for $75$ piano students in this sample who practiced at least $50$ minutes on average each week.

| | Percentage of correct notes the students played during a recital | | | |
|---|---|---|---|---|
| **Average number of minutes practiced each week** | **Less than $50\%$** | **$50\%$ to $80\%$** | **Greater than $80\%$** | **Total** |
| $50$ **to** $150$ | $7$ | $28$ | $5$ | $40$ |
| $151$ **or more** | $3$ | $11$ | $21$ | $35$ |
| **Total** | $10$ | $39$ | $26$ | $75$ |

Which of the following is the largest population to which the results of the study can be generalized?

**Answer**

A. All piano students who were in their second year with their conservatory

B. All piano students in the region

C. All piano students in the region who were in their second year with their conservatory

D. The $75$ piano students in the sample who practiced at least $50$ minutes on average each week

**Correct Answer: C**

**Rationale**

Choice C is correct. It's given that the statistician selected a sample of $85$ piano students at random from all piano students in a certain region who were in their second year with their conservatory. It's also given that the table shows the information for $75$ piano students in the sample who practiced at least $50$ minutes on average each week. Selecting a sample at random from a population allows the results of the study to be generalized to that population, but not beyond it. Therefore, the largest population to which the results of the study can be generalized is the population of all piano students in the region who were in their second year with their conservatory.

Choice A is incorrect. The results can't be generalized to all piano students who were in their second year with their conservatory since only students in a certain region were sampled.

Choice B is incorrect. The results can't be generalized to all piano students in the region since only students in their second year with their conservatory were sampled.

Choice D is incorrect. The results of the study can be generalized beyond the sample itself to the population from which it was drawn.

# Question ID: aa43b41f

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Evaluating statistical claims: Observational studies and experiments | Hard |

**Question**

Near the end of a US cable news show, the host invited viewers to respond to a poll on the show's website that asked, "Do you support the new federal policy discussed during the show?" At the end of the show, the host reported that 28% responded "Yes," and 70% responded "No." Which of the following best explains why the results are unlikely to represent the sentiments of the population of the United States?

**Answer**

A. The percentages do not add up to 100%, so any possible conclusions from the poll are invalid.

B. Those who responded to the poll were not a random sample of the population of the United States.

C. There were not 50% "Yes" responses and 50% "No" responses.

D. The show did not allow viewers enough time to respond to the poll.

**Correct Answer: B**

**Rationale**

Choice B is correct. In order for the poll results from a sample of a population to represent the entire population, the sample must be representative of the population. A sample that is randomly selected from a population is more likely than a sample of the type described to represent the population. In this case, the people who responded were people with access to cable television and websites, which aren't accessible to the entire population. Moreover, the people who responded also chose to watch the show and respond to the poll. The people who made these choices aren't representative of the entire population of the United States because they were not a random sample of the population of the United States.

Choices A, C, and D are incorrect because they present reasons unrelated to whether the sample is representative of the population of the United States.

## Question ID: 5dc386fb

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Probability and conditional probability | Hard |

**Question**

The table below shows the distribution of US states according to whether they have a state-level sales tax and a state-level income tax.

2013 State-Level Taxes

| | State sales tax | No state sales tax |
|---|---|---|
| State income tax | 39 | 4 |
| No state income tax | 6 | 1 |

To the nearest tenth of a percent, what percent of states with a state-level sales tax do not have a state-level income tax?

**Answer**

A. 6.0%

B. 12.0%

C. 13.3%

D. 14.0%

**Correct Answer: C**

**Rationale**

Choice C is correct. The sum of the number of states with a state-level sales tax is $39 + 6 = 45$. Of these states, 6 don't have a state-level income tax. Therefore, $\frac{6}{45} = 0.1333...$, or about 13.3%, of states with a state-level sales tax don't have a state-level income tax.

Choice A is incorrect. This is the number of states that have a state-level sales tax and no state-level income tax. Choice B is incorrect. This is the percent of states that have a state-level sales tax and no state-level income tax. Choice D is incorrect. This is the percent of states that have no state-level income tax.

## Question ID: 014c47ab

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Probability and conditional probability | Hard |

**Question**

| | Site A | Site B | Total |
|---|---|---|---|
| **Tulip** | $35$ | $15$ | $50$ |
| **Daffodil** | $31$ | $21$ | $52$ |
| **Total** | $66$ | $36$ | $102$ |

The table shows the distribution of two types of flowers at two different sites. If a flower represented in the table is selected at random, what is the probability of selecting a flower from site A, given that the flower is a tulip? (Express your answer as a decimal or fraction, not as a percent.)

**Correct Answer: 0.7, 7/10**

**Rationale**

The correct answer is $\frac{35}{50}$. Based on the table, there are a total of $50$ tulips, and $35$ of these tulips are from site A. The probability of selecting at random a flower from site A, given that the flower is a tulip, is equal to the number of tulips from site A divided by the total number of tulips, which can be written as $\frac{35}{50}$, or $\frac{7}{10}$. Note that 35/50, 7/10, and .7 are examples of ways to enter a correct answer.

## Question ID: 98958ae8

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard |

**Question**

Data set A consists of the heights of $75$ objects and has a mean of $25$ meters. Data set B consists of the heights of $50$ objects and has a mean of $65$ meters. Data set C consists of the heights of the $125$ objects from data sets A and B. What is the mean, in meters, of data set C?

**Correct Answer: 41**

**Rationale**

The correct answer is $41$. The mean of a data set is computed by dividing the sum of the values in the data set by the number of values in the data set. It's given that data set A consists of the heights of $75$ objects and has a mean of $25$ meters. This can be represented by the equation $\frac{x}{75} = 25$, where $x$ represents the sum of the heights of the objects, in meters, in data set A. Multiplying both sides of this equation by $75$ yields $x = 75(25)$, or $x = 1{,}875$ meters. Therefore, the sum of the heights of the objects in data set A is $1{,}875$ meters. It's also given that data set B consists of the heights of $50$ objects and has a mean of $65$ meters. This can be represented by the equation $\frac{y}{50} = 65$, where $y$ represents the sum of the heights of the objects, in meters, in data set B. Multiplying both sides of this equation by $50$ yields $y = 50(65)$, or $y = 3{,}250$ meters. Therefore, the sum of the heights of the objects in data set B is $3{,}250$ meters. Since it's given that data set C consists of the heights of the $125$ objects from data sets A and B, it follows that the mean of data set C is the sum of the heights of the objects, in meters, in data sets A and B divided by the number of objects represented in data sets A and B, or $\frac{1{,}875+3{,}250}{125}$, which is equivalent to $41$ meters. Therefore, the mean, in meters, of data set C is $41$.

## Question ID: 623dbebb

| Assessment | Test | Domain | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Problem-Solving and Data Analysis | Percentages | Hard |

**Question**

A reseller buys certain books for a purchase price of $5.00$ dollars each and then marks them each for sale at a consumer price that is $270\%$ of the purchase price. After $4$ months, any remaining books not yet sold are marked at a discounted price that is $70\%$ off the consumer price. What is the discounted price of each of the remaining books, in dollars?

**Correct Answer: 4.05, 81/20**

**Rationale**

The correct answer is $4.05$. It's given that the purchase price for certain books is $5.00$ dollars each. It's also given that each book is marked for sale at a consumer price that is $270\%$ of the purchase price. Since the consumer price is $270\%$ of the purchase price of $5.00$ dollars, it follows that the consumer price is $(2.7)(5.00)$, or $13.50$, dollars. It's given that after $4$ months, any remaining books are discounted at $70\%$ off the consumer price. Thus, the discount amount is $(0.7)(13.50)$, or $9.45$, dollars. Subtracting the discount amount from the consumer price gives the discounted price of each of the remaining books: $13.50 - 9.45 = 4.05$. Note that 4.05 and 81/20 are examples of ways to enter a correct answer.

## Question ID: 2e92cc21

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Percentages | Hard |

**Question**

The number $a$ is $110\%$ greater than the number $b$. The number $b$ is $90\%$ less than $47$. What is the value of $a$?

**Correct Answer: 9.87, 987/100**

**Rationale**

The correct answer is $9.87$. It's given that the number $a$ is $110\%$ greater than the number $b$. It follows that $a = \left(1 + \frac{110}{100}\right)b$, or $a = 2.1b$. It's also given that the number $b$ is $90\%$ less than $47$. It follows that $b = \left(1 - \frac{90}{100}\right)(47)$, or $b = 0.1(47)$, which yields $b = 4.7$. Substituting $4.7$ for $b$ in the equation $a = 2.1b$ yields $a = 2.1(4.7)$, which is equivalent to $a = 9.87$. Therefore, the value of $a$ is $9.87$.

## Question ID: 4a422e3e

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Evaluating statistical claims: Observational studies and experiments | Hard |

**Question**

To determine the mean number of children per household in a community, Tabitha surveyed 20 families at a playground. For the 20 families surveyed, the mean number of children per household was 2.4. Which of the following statements must be true?

**Answer**

A. The mean number of children per household in the community is 2.4.

B. A determination about the mean number of children per household in the community should not be made because the sample size is too small.

C. The sampling method is flawed and may produce a biased estimate of the mean number of children per household in the community.

D. The sampling method is not flawed and is likely to produce an unbiased estimate of the mean number of children per household in the community.

**Correct Answer: C**

**Rationale**

Choice C is correct. In order to use a sample mean to estimate the mean for a population, the sample must be representative of the population (for example, a simple random sample). In this case, Tabitha surveyed 20 families in a playground. Families in the playground are more likely to have children than other households in the community. Therefore, the sample isn't representative of the population. Hence, the sampling method is flawed and may produce a biased estimate.

Choices A and D are incorrect because they incorrectly assume the sampling method is unbiased. Choice B is incorrect because a sample of size 20 could be large enough to make an estimate if the sample had been representative of all the families in the community.

# Question ID: f496d2c0

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Probability and conditional probability | Hard |

**Question**

The table shows the distribution of rooms in a certain facility by seating capacity.

| Seating capacity | Proportion |
|---|---|
| Less than $18$ seats | $26\%$ |
| $18–40$ seats | $21\%$ |
| $41–65$ seats | $29\%$ |
| Greater than $65$ seats | $24\%$ |

If a room in this facility is selected at random, which of the following is closest to the probability of selecting a room that has a seating capacity greater than $65$ seats, given that the room has a seating capacity of at least $18$ seats?

**Answer**

A. $0.24$

B. $0.32$

C. $0.50$

D. $0.92$

**Correct Answer: B**

**Rationale**

Choice B is correct. It's given that the proportion of rooms with less than $18$ seats is $26\%$, which is equivalent to $0.26$. It's also given that the proportion of rooms with $18–40$ seats is $21\%$, or $0.21$; that the proportion of rooms with $41–65$ seats is $29\%$, or $0.29$; and that the proportion of rooms with greater than $65$ seats is $24\%$, or $0.24$. The probability of selecting a room that has a seating capacity greater than $65$ seats given that the room has a seating capacity of at least $18$ seats is equal to the proportion of rooms that have a seating capacity of greater than $65$ seats divided by the proportion of rooms that have a seating capacity of at least $18$ seats. Rooms with a seating capacity of at least $18$ seats include the categories $18–40$ seats, $41–65$ seats, and greater than $65$ seats, so the total proportion for this condition can be written as $0.21 + 0.29 + 0.24$, or $0.74$. Therefore, the probability of selecting a room that has a seating capacity greater than $65$ seats given that it has a seating capacity of at least $18$ seats is $\frac{0.24}{0.74}$, or approximately $0.324$. Of the given choices, $0.32$ is closest to $0.324$.

Choice A is incorrect. This is the probability of selecting a room that has a seating capacity greater than $65$ seats from all rooms in the facility, not just rooms that meet the condition of having a seating capacity of at least $18$ seats.

Choice C is incorrect and may result from conceptual or calculation errors.

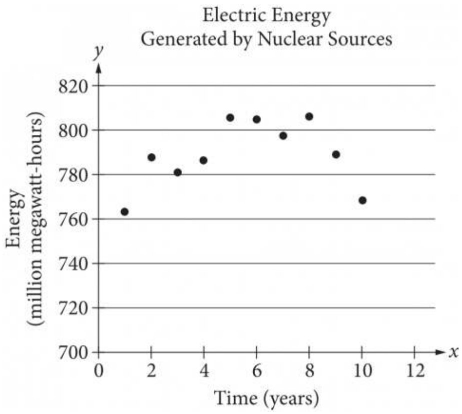
Choice D is incorrect and may result from conceptual or calculation errors.

# Question ID: e821a26d

| Assessment | Test | Domain | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Problem-Solving and Data Analysis | Two-variable data: Models and scatterplots | Hard |

**Question**

The scatterplot below shows the amount of electric energy generated, in millions of megawatt-hours, by nuclear sources over a 10-year period.




Of the following equations, which best models the data in the scatterplot?

**Answer**

A. $y = 1.674x^2 + 19.76x - 745.73$

B. $y = -1.674x^2 - 19.76x - 745.73$

C. $y = 1.674x^2 + 19.76x + 745.73$

D. $y = -1.674x^2 + 19.76x + 745.73$

**Correct Answer: D**

**Rationale**

Choice D is correct. The data in the scatterplot roughly fall in the shape of a downward-opening parabola; therefore, the coefficient for the $x^2$ term must be negative. Based on the location of the data points, the y-intercept of the parabola should be somewhere between 740 and 760. Therefore, of the equations given, the best model is $y = -1.674x^2 + 19.76x + 745.73$.

Choices A and C are incorrect. The positive coefficient of the $x^2$ term means that these equations each define upward-opening parabolas, whereas a parabola that fits the data in the scatterplot must open downward. Choice B is incorrect because it defines a parabola with a y-intercept that has a negative y-coordinate, whereas a parabola that fits the data in the scatterplot must have a y-intercept with a positive y-coordinate.

## Question ID: 9d935bd8

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard |

**Question**

Percent of Residents Who Earned a Bachelor's Degree or Higher

| State | Percent of residents |
|---|---|
| State A | 21.9% |
| State B | 27.9% |
| State C | 25.9% |
| State D | 19.5% |
| State E | 30.1% |
| State F | 36.4% |
| State G | 35.5% |

A survey was given to residents of all 50 states asking if they had earned a bachelor's degree or higher. The results from 7 of the states are given in the table above. The median percent of residents who earned a bachelor's degree or higher for all 50 states was 26.95%. What is the difference between the median percent of residents who earned a bachelor's degree or higher for these 7 states and the median for all 50 states?

**Answer**

A. 0.05%

B. 0.95%

C. 1.22%

D. 7.45%

**Correct Answer: B**

**Rationale**

Choice B is correct. The median of a set of numbers is the middle value of the set values when ordered from least to greatest. If the percents in the table are ordered from least to greatest, the middle value is 27.9%. The difference between 27.9% and 26.95% is 0.95%.

Choice A is incorrect and may be the result of calculation errors or not finding the median of the data in the table correctly. Choice C is incorrect and may be the result of finding the mean instead of the median. Choice D is incorrect and may be the result of using the middle value of the unordered list.

# Question ID: 8c5dbd3e

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Percentages | Hard |

**Question**

The number $w$ is $110\%$ greater than the number $z$. The number $z$ is $55\%$ less than $50$. What is the value of $w$?

**Correct Answer: 189/4, 47.25**

**Rationale**

The correct answer is $47.25$. It's given that the number $w$ is $110\%$ greater than the number $z$. It follows that $w = \left(1 + \frac{110}{100}\right)z$, or $w = 2.1z$. It's also given that the number $z$ is $55\%$ less than $50$. It follows that $z = \left(1 - \frac{55}{100}\right)(50)$, or $z = 0.45(50)$, which yields $z = 22.5$. Substituting $22.5$ for $z$ in the equation $w = 2.1z$ yields $w = 2.1(22.5)$, which is equivalent to $w = 47.25$. Therefore, the value of $w$ is $47.25$. Note that 47.25 and 189/4 are examples of ways to enter a correct answer.

## Question ID: da978ee6

| Assessment | Test | Domain | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Problem-Solving and Data Analysis | Inference from sample statistics and margin of error | Hard |

**Question**

A researcher surveyed a random sample of $1,020$ people from a certain city to estimate the percentage of people who support a recently announced proposal. From the survey, the researcher estimated that $86\%$ of people from the city support this proposal, with an associated margin of error of $2.13\%$. The researcher repeated the survey with a random sample of people from the city that was double the original sample size. Assuming the margins of error were calculated in the same way, which of the following is true about the margin of error associated with the estimate from the larger sample?

**Answer**

A. The margin of error is between $2.13\%$ and $3.13\%$.

B. The margin of error is between $4.13\%$ and $5.13\%$.

C. The margin of error is greater than $5.13\%$.

D. The margin of error is less than $2.13\%$.

**Correct Answer: D**

**Rationale**

Choice D is correct. It's given that from the initial survey, the margin of error associated with the estimate was $2.13\%$. A larger sample size generally leads to a smaller margin of error. Therefore, assuming the margins of error were calculated in the same way, the margin of error associated with the estimate from the larger sample is less than $2.13\%$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

## Question ID: 9ba3e283

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Inference from sample statistics and margin of error | Hard |

**Question**

In State X, Mr. Camp's eighth-grade class consisting of 26 students was surveyed and 34.6 percent of the students reported that they had at least two siblings. The average eighth-grade class size in the state is 26. If the students in Mr. Camp's class are representative of students in the state's eighth-grade classes and there are 1,800 eighth-grade classes in the state, which of the following best estimates the number of eighth-grade students in the state who have fewer than two siblings?

**Answer**

A. 16,200

B. 23,400

C. 30,600

D. 46,800

**Correct Answer: C**

**Rationale**

Choice C is correct. It is given that 34.6% of 26 students in Mr. Camp's class reported that they had at least two siblings. Since 34.6% of 26 is 8.996, there must have been 9 students in the class who reported having at least two siblings and 17 students who reported that they had fewer than two siblings. It is also given that the average eighth-grade class size in the state is 26 and that Mr. Camp's class is representative of all eighth-grade classes in the state. This means that in each eighth-grade class in the state there are about 17 students who have fewer than two siblings. Therefore, the best estimate of the number of eighth-grade students in the state who have fewer than two siblings is 17 × (number of eighth-grade classes in the state), or $17 \times 1{,}800 = 30{,}600$.

Choice A is incorrect because 16,200 is the best estimate for the number of eighth-grade students in the state who have at least, not fewer than, two siblings. Choice B is incorrect because 23,400 is half of the estimated total number of eighth-grade students in the state; however, since the students in Mr. Camp's class are representative of students in the eighth-grade classes in the state and more than half of the students in Mr. Camp's class have fewer than two siblings, more than half of the students in each eighth-grade class in the state have fewer than two siblings, too. Choice D is incorrect because 46,800 is the estimated total number of eighth-grade students in the state.

# Question ID: 6feae9c3

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Percentages | Hard |

**Question**

At a recreation center, visitors used the exercise room for a total of $t$ hours in July. At the same recreation center, visitors used the exercise room for $3.49t$ hours in August. What is the percent increase in the number of hours visitors used the exercise room from July to August?

**Answer**

A. $2.49\%$

B. $3.49\%$

C. $249\%$

D. $349\%$

**Correct Answer: C**

**Rationale**

Choice C is correct. It’s given that the total number of hours visitors used the exercise room increased from $t$ hours in July to $3.49t$ hours in August. It follows that the number of hours increased by $3.49t - t$, or $2.49t$. Let $p\%$ represent the percent increase in the exercise room use from July to August, or the percentage that $2.49t$ is of $t$. The value of $p$ can be found using the proportion $\frac{2.49t}{t} = \frac{p}{100}$, or $2.49 = \frac{p}{100}$. Multiplying both sides of this equation by $100$ yields $p = 249$. Therefore, the percent increase in the number of hours visitors used the exercise room from July to August is $249\%$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. The exercise room use in August is $349\%$ of the exercise room use in July, but this doesn't represent the percent increase from July to August.

## Question ID: 89f20d9e

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Probability and conditional probability | Hard |

**Question**

The table summarizes the distribution of age and assigned group for $90$ participants in a study.

| | $0–9$ years | $10–19$ years | $20+$ years | Total |
|---|---|---|---|---|
| **Group A** | $5$ | $17$ | $8$ | $30$ |
| **Group B** | $6$ | $8$ | $16$ | $30$ |
| **Group C** | $19$ | $5$ | $6$ | $30$ |
| **Total** | $30$ | $30$ | $30$ | $90$ |

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least 10 years of age?

**Answer**

A. $\frac{5}{18}$

B. $\frac{5}{12}$

C. $\frac{17}{30}$

D. $\frac{5}{6}$

**Correct Answer: B**

**Rationale**

Choice B is correct. Since the participant will be selected at random, the probability of selecting a participant from group A, given that the participant is at least $10$ years of age, is equal to the number of participants from group A who are at least $10$ years of age divided by the total number of participants who are at least $10$ years of age. Based on the table, in group A, there are $17$ participants who are $10–19$ years of age and $8$ participants who are $20+$ years of age. Therefore, there are a total of $17 + 8$, or $25$, participants in group A who are at least $10$ years of age. Based on the table, of the total number of participants, there are $30$ participants who are $10–19$ years of age and $30$ participants who are $20+$ years of age. Therefore, a total of $30 + 30$, or $60$, of the participants are at least $10$ years of age. Thus, the probability of selecting a participant from group A, given that the participant is at least $10$ years of age, is $\frac{25}{60}$, or $\frac{5}{12}$.

Choice A is incorrect. This is the number of participants from group A who are at least $10$ years of age divided by the total number of participants, rather than divided by the number of participants who are at least $10$ years of age.

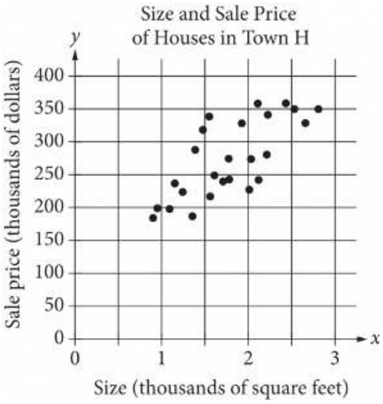
Choice C is incorrect. This is the probability of randomly selecting a participant from group A, given that the participant is $10–19$ years of age, rather than given that the participant is at least $10$ years of age.

Choice D is incorrect. This is the probability of randomly selecting a participant who is at least $10$ years of age, given that the participant is in group A.

# Question ID: 79137c1b

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Two-variable data: Models and scatterplots | Hard |

**Question**




The scatterplot above shows the size x and the sale price y of 25 houses for sale in Town H. Which of the following could be an equation for a line of best fit for the data?

**Answer**

A. $y = 200x + 100$

B. $y = 100x + 100$

C. $y = 50x + 100$

D. $y = 100x$

**Correct Answer: B**

**Rationale**

Choice B is correct. From the shape of the cluster of points, the line of best fit should pass roughly through the points $(1, 200)$ and $(2.5, 350)$. Therefore, these two points can be used to find an approximate equation for the line of best fit. The slope of this line of best fit is therefore $\frac{y_2 - y_1}{x_2 - x_1} = \frac{350 - 200}{2.5 - 1}$, or 100. The equation for the line of best fit, in slope-intercept form, is $y = 100x + b$ for some value of b. Using the point $(1, 200)$, 1 can be substituted for x and 200 can be substituted for y: $200 = 100(1) + b$, or $b = 100$. Substituting this value into the slope-intercept form of the equation gives $y = 100x + 100$.

Choice A is incorrect. The line defined by $y = 200x + 100$ passes through the points $(1, 300)$ and $(2, 500)$, both of which are well above the cluster of points, so it cannot be a line of best fit. Choice C is incorrect. The line defined by $y = 50x + 100$ passes through the points $(1, 150)$ and $(2, 200)$, both of which lie at the bottom of the cluster of points, so it cannot be a line of best fit. Choice D is incorrect and may result from correctly calculating the slope of a line of best fit but incorrectly assuming the y-intercept is at $(0, 0)$.

## Question ID: a51e6f93

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Percentages | Hard |

**Question**

$0.1152$ is $0.08\%$ of the number $b$. The number $b$ is $25\%$ of the number $c$. What is the value of $c$?

**Correct Answer: 576**

**Rationale**

The correct answer is $576$. It's given that $0.1152$ is $0.08\%$ of the number $b$. It follows that $0.1152 = \frac{0.08}{100}b$, or $0.1152 = 0.0008b$. Dividing both sides of this equation by $0.0008$ yields $144 = b$. It's also given that the number $b$ is $25\%$ of the number $c$. It follows that $b = \frac{25}{100}c$, or $b = 0.25c$. Substituting $144$ for $b$ in this equation yields $144 = 0.25c$. Dividing both sides of this equation by $0.25$ yields $576 = c$. Therefore, the value of $c$ is $576$.

## Question ID: 54d93874

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard |

**Question**

| | Masses (kilograms) | | | | | |
|---|---|---|---|---|---|---|
| **Andrew** | 2.4 | 2.5 | 3.6 | 3.1 | 2.5 | 2.7 |
| **Maria** | x | 3.1 | 2.7 | 2.9 | 3.3 | 2.8 |

Andrew and Maria each collected six rocks, and the masses of the rocks are shown in the table above. The mean of the masses of the rocks Maria collected is 0.1 kilogram greater than the mean of the masses of the rocks Andrew collected. What is the value of x ?

**Rationale**

The correct answer is 2.6. Since the mean of a set of numbers can be found by adding the numbers together and dividing by how many numbers there are in the set, the mean mass, in kilograms, of the rocks Andrew collected is $\frac{2.4+2.5+3.6+3.1+2.5+2.7}{6}=\frac{16.8}{6}$, or 2.8.

Since the mean mass of the rocks Maria collected is 0.1 kilogram greater than the mean mass of rocks Andrew collected, the mean mass of the rocks Maria collected is $2.8+0.1=2.9$ kilograms. The value of x can be found by writing an equation for finding the mean: $\frac{x+3.1+2.7+2.9+3.3+2.8}{6}=2.9$. Solving this equation gives $x=2.6$. Note that 2.6 and 13/5 are examples of ways to enter a correct answer.

## Question ID: af142f8d

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Two-variable data: Models and scatterplots | Hard |

**Question**

| | Amount invested | Balance increase |
|---|---|---|
| Account A | \$500 | 6% annual interest |
| Account B | \$1,000 | \$25 per year |

Two investments were made as shown in the table above. The interest in Account A is compounded once per year. Which of the following is true about the investments?

**Answer**

A. Account A always earns more money per year than Account B.

B. Account A always earns less money per year than Account B.

C. Account A earns more money per year than Account B at first but eventually earns less money per year.

D. Account A earns less money per year than Account B at first but eventually earns more money per year.

**Correct Answer: A**

**Rationale**

Choice A is correct. Account A starts with \$500 and earns interest at 6% per year, so in the first year Account A earns (500)(0.06) = \$30, which is greater than the \$25 that Account B earns that year. Compounding interest can be modeled by an increasing exponential function, so each year Account A will earn more money than it did the previous year. Therefore, each year Account A earns at least \$30 in interest. Since Account B always earns \$25 each year, Account A always earns more money per year than Account B.

Choices B and D are incorrect. Account A earns \$30 in the first year, which is greater than the \$25 Account B earns in the first year. Therefore, neither the statement that Account A always earns less money per year than Account B nor the statement that Account A earns less money than Account B at first can be true. Choice C is incorrect. Since compounding interest can be modeled by an increasing exponential function, each year Account A will earn more money than it did the previous year. Therefore, Account A always earns at least \$30 per year, which is more than the \$25 per year that Account B earns.

## Question ID: 8213b1b3

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Percentages | Hard |

**Question**

According to a set of standards, a certain type of substance can contain a maximum of $0.001\%$ phosphorus by mass. If a sample of this substance has a mass of $140$ grams, what is the maximum mass, in grams, of phosphorus the sample can contain to meet these standards?

**Correct Answer: .0014**

**Rationale**

The correct answer is $.0014$. It's given that a certain type of substance can contain a maximum of $0.001\%$ phosphorus by mass to meet a set of standards. If a sample of the substance has a mass of $140$ grams, it follows that the maximum mass, in grams, of phosphorus the sample can contain to meet the standards is $0.001\%$ of $140$, or $\frac{0.001}{100}(140)$, which is equivalent to $(0.00001)(140)$, or $0.0014$. Note that .0014 and 0.001 are examples of ways to enter a correct answer.

## Question ID: 34f8cd89

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Percentages | Hard |

**Question**

$37\%$ of the items in a box are green. Of those, $37\%$ are also rectangular. Of the green rectangular items, $42\%$ are also metal. Which of the following is closest to the percentage of the items in the box that are <u>not</u> rectangular green metal items?

**Answer**

A. $1.16\%$

B. $57.50\%$

C. $94.25\%$

D. $98.84\%$

**Correct Answer: C**

**Rationale**

Choice C is correct. It's given that $37\%$ of the items in a box are green. Let $x$ represent the total number of items in the box. It follows that $\frac{37}{100}x$, or $0.37x$, items in the box are green. It's also given that of those, $37\%$ are also rectangular. Therefore, $\frac{37}{100}(0.37x)$, or $0.1369x$, items in the box are green rectangular items. It's also given that of the green rectangular items, $42\%$ are also metal. Therefore, $\frac{42}{100}(0.1369x)$, or $0.057498x$, items in the box are rectangular green metal items. The number of the items in the box that are not rectangular green metal items is the total number of items in the box minus the number of rectangular green metal items in the box. Therefore, the number of items in the box that are not rectangular green metal items is $x - 0.057498x$, or $0.942502x$. The percentage of items in the box that are not rectangular green metal items is the percentage that $0.942502x$ is of $x$. If $p\%$ represents this percentage, the value of $p$ is $100\left(\frac{0.942502x}{x}\right)$, or $94.2502$. Of the given choices, $94.25\%$ is closest to the percentage of items in the box that are not rectangular green metal items.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

# Question ID: 6fca0144

| Assessment | Test | Domain | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Problem-Solving and Data Analysis | Evaluating statistical claims: Observational studies and experiments | Hard |

**Question**

For a baobab tree habitat in South Africa, a scientist randomly selected $50$ baobab trees that were $17$ years old and randomly assigned them to two groups. Each group was subjected to a different watering pattern for $2$ consecutive years to observe whether the watering pattern affects the trees' growth rate. Based on the design of the study, what is the largest group to which these results can be applied?

**Answer**

A. All the $50$ baobab trees that were selected in this habitat

B. All the baobab trees that were $19$ years old in this habitat

C. All the baobab trees that were $17$ years old in South Africa

D. All the baobab trees that were $17$ years old in this habitat

**Correct Answer: D**

**Rationale**

Choice D is correct. When a study uses a randomly selected sample, the largest group to which the results of the study can be applied is the population from which the sample was selected. It's given that the scientist randomly selected the trees from the baobab trees in a certain habitat that were $17$ years old. Therefore, the largest group to which the results of this study can be applied is all the baobab trees that were $17$ years old in this habitat.

Choice A is incorrect. Since the sample was randomly selected from a population, the results can be applied to a larger group than the sample.

Choice B is incorrect. The sample was selected from a population of baobab trees that were $17$ years old, not $19$ years old.

Choice C is incorrect. The sample was selected from a certain tree habitat in South Africa, not from all the baobab trees that were $17$ years old in South Africa.

## Question ID: 20b69297

| Assessment | Test | Domain | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Problem-Solving and Data Analysis | Ratios, rates, proportional relationships, and units | Hard |

**Question**

Anita created a batch of green paint by mixing 2 ounces of blue paint with 3 ounces of yellow paint. She must mix a second batch using the same ratio of blue and yellow paint as the first batch. If she uses 5 ounces of blue paint for the second batch, how much yellow paint should Anita use?

**Answer**

A. Exactly 5 ounces

B. 3 ounces more than the amount of yellow paint used in the first batch

C. 1.5 times the amount of yellow paint used in the first batch

D. 1.5 times the amount of blue paint used in the second batch

**Correct Answer: D**

**Rationale**

Choice D is correct. It's given that Anita used a ratio of 2 ounces of blue paint to 3 ounces of yellow paint for the first batch. For any batch of paint that uses the same ratio, the amount of yellow paint used will be $\frac{3}{2}$, or 1.5, times the amount of blue paint used in the batch. Therefore, the amount of yellow paint Anita will use in the second batch will be 1.5 times the amount of blue paint used in the second batch.

Alternate approach: It's given that Anita used a ratio of 2 ounces of blue paint to 3 ounces of yellow paint for the first batch and that she will use 5 ounces of blue paint for the second batch. A proportion can be set up to solve for x, the amount of yellow paint she will use for the second batch: $\frac{2}{3} = \frac{5}{x}$. Multiplying both sides of this equation by 3 yields $2 = \frac{15}{x}$, and multiplying both sides of this equation by x yields $2x = 15$. Dividing both sides of this equation by 2 yields $x = 7.5$. Since Anita will use 7.5 ounces of yellow paint for the second batch, this is $\frac{7.5}{5} = 1.5$ times the amount of blue paint (5 ounces) used in the second batch.

Choices A, B, and C are incorrect and may result from incorrectly interpreting the ratio of blue paint to yellow paint used.

## Question ID: 94237701

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard |

**Question**

For a certain computer game, individuals receive an integer score that ranges from 2 through 10. The table below shows the frequency distribution of the scores of the 9 players in group A and the 11 players in group B.

| Score | Score Frequencies | |
|---|---|---|
| | Group A | Group B |
| **2** | 1 | 0 |
| **3** | 1 | 0 |
| **4** | 2 | 0 |
| **5** | 1 | 4 |
| **6** | 3 | 2 |
| **7** | 0 | 0 |
| **8** | 0 | 2 |
| **9** | 1 | 1 |
| **10** | 0 | 2 |
| **Total** | 9 | 11 |

The median of the scores for group B is how much greater than the median of the scores for group A?

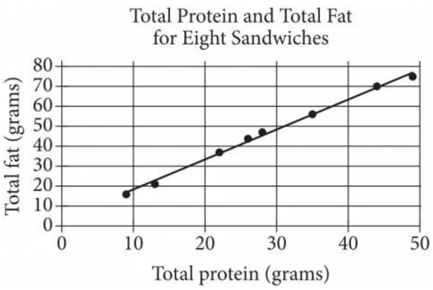
**Rationale**

The correct answer is 1. When there are an odd number of values in a data set, the median of the data set is the middle number when the data values are ordered from least to greatest. The scores for group A, ordered from least to greatest, are 2, 3, 4, 4, 5, 6, 6, 6, and 9. The median of the scores for group A is therefore 5. The scores for group B, ordered from least to greatest, are 5, 5, 5, 5, 6, 6, 8, 8, 9, 10, and 10. The median of the scores for group B is therefore 6. The median score for group B is $6-5=1$ more than the median score for group A.

# Question ID: 9d95e7ad

| Assessment | Test | Domain | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Problem-Solving and Data Analysis | Two-variable data: Models and scatterplots | Hard |

**Question**




The scatterplot above shows the numbers of grams of both total protein and total fat for eight sandwiches on a restaurant menu. The line of best fit for the data is also shown. According to the line of best fit, which of the following is closest to the predicted increase in total fat, in grams, for every increase of 1 gram in total protein?

**Answer**

A. 2.5

B. 2.0

C. 1.5

D. 1.0

**Correct Answer: C**

**Rationale**

Choice C is correct. The predicted increase in total fat, in grams, for every increase of 1 gram in total protein is represented by the slope of the line of best fit. Any two points on the line can be used to calculate the slope of the line as the change in total fat over the change in total protein. For instance, it can be estimated that the points $(20,34)$ and $(30,48)$ are on the line of best fit, and the slope of the line that passes through them is $\frac{48-34}{30-20}=\frac{14}{10}$, or 1.4. Of the choices given, 1.5 is the closest to the slope of the line of best fit.

Choices A, B, and D are incorrect and may be the result of incorrectly finding ordered pairs that lie on the line of best fit or of incorrectly calculating the slope.

## Question ID: 11b06e35

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Ratios, rates, proportional relationships, and units | Hard |

**Question**

The density of a certain solid substance is $813$ kilograms per cubic meter. A sample of this substance is in the shape of a cube, where each edge has a length of $0.60$ meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

**Answer**

A. $176$

B. $488$

C. $1{,}355$

D. $3{,}764$

**Correct Answer: A**

**Rationale**

Choice A is correct. It's given that the sample is in the shape of a cube with edge lengths of $0.60$ meters. Therefore, the volume of the sample is $0.60^3$, or $0.216$, cubic meters. It's also given that the sample has a density of $813$ kilograms per $1$ cubic meter. Therefore, the mass of this sample is $(0.216 \text{ cubic meters})\left(\frac{813 \text{ kilograms}}{1 \text{ cubic meter}}\right)$, or $175.608$ kilograms. Rounding this mass to the nearest whole number gives $176$ kilograms. Therefore, to the nearest whole number, the mass, in kilograms, of this sample is $176$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

# Question ID: cf2be18d

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | One-variable data: Distributions and measures of center and spread | Hard |

**Question**

$$a, 26, 29, b, 31, 47, c$$

For the given data set, the data values are listed in ascending order, where $a$, $b$, and $c$ are constants. For this data set, the mean is $36$, the median is $29$, and the range is $72$. What is the value of $c$?

**Answer**

A. $54$

B. $72$

C. $81$

D. $98$

**Correct Answer: C**

**Rationale**

Choice C is correct. It's given that the data set containing $a$, $26$, $29$, $b$, $31$, $47$, and $c$ is listed in ascending order and has a mean of $36$, a median of $29$, and a range of $72$. Since there are $7$ values in the given data set, it follows that the median is the value in the fourth position. It's given that $b$ is in the fourth position in the given data set, so it follows that $b = 29$. Since the mean of the data set is $36$, it follows that $\frac{(a+26+29+29+31+47+c)}{7} = 36$. Multiplying both sides of this equation by $7$ yields $a + 26 + 29 + 29 + 31 + 47 + c = 252$, which is equivalent to $a + c + 162 = 252$. Subtracting $162$ from both sides of this equation yields $a + c = 90$. The range is the difference between the greatest and least values in a data set. Since $c$ is the greatest value and $a$ is the least value in the given data set, it follows that $c - a = 72$, or $-a + c = 72$. Adding the equations $a + c = 90$ and $-a + c = 72$ yields $2c = 162$. Dividing both sides of this equation by $2$ yields $c = 81$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

## Question ID: d6456c7a

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Ratios, rates, proportional relationships, and units | Hard |

**Question**

A certain park has an area of $11{,}863{,}808$ square yards. What is the area, in square miles, of this park? $(1 \text{ mile} = 1{,}760 \text{ yards})$

**Answer**

A. $1.96$

B. $3.83$

C. $3{,}444.39$

D. $6{,}740.8$

**Correct Answer: B**

**Rationale**

Choice B is correct. Since $1$ mile is equal to $1{,}760$ yards, $1$ square mile is equal to $1{,}760^2$, or $3{,}097{,}600$, square yards. It's given that the park has an area of $11{,}863{,}808$ square yards. Therefore, the park has an area of $(11{,}863{,}808 \text{ square yards})\left(\frac{1 \text{ square mile}}{3{,}097{,}600 \text{ square yards}}\right)$, or $\frac{11{,}863{,}808}{3{,}097{,}600}$ square miles. Thus, the area, in square miles, of the park is $3.83$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the square root of the area of the park in square yards, not the area of the park in square miles.

Choice D is incorrect and may result from converting $11{,}863{,}808$ yards to miles, rather than converting $11{,}863{,}808$ square yards to square miles.

## Question ID: 98818acf

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Percentages | Hard |

**Question**

The number of audiobooks in a library increased by $131\%$ from last year to this year. The number of audiobooks in the library last year was $n$, and the number of audiobooks in the library this year is $bn$, where $b$ and $n$ are constants. What is the value of $b$?

**Correct Answer: 2.31**

**Rationale**

The correct answer is $2.31$. It's given that the number of audiobooks in the library increased by $131\%$ from last year to this year. Converting $131\%$ to a decimal yields $1.31$. It's also given that the number of audiobooks in the library last year was $n$, where $n$ is a constant. It follows that the number of audiobooks in the library this year can be represented by $n$ plus $131\%$ of $n$, or $n + 1.31n$. Combining like terms in this expression yields $2.31n$. Since the number of audiobooks in the library this year is represented as $bn$, where $b$ is a constant, it follows that the value of $b$ is $2.31$.